

KEVIN O'DONNELL

Los Angeles, CA | kao2160@columbia.edu | (310) 560-4009 | kevinod.com/Projects

EDUCATION

Columbia University

NY, NY | Sep 2022 - Dec 2024

Master of Science in Biomedical Engineering. Concentration in Robotics and Control. GPA: 3.5/4.0

Relevant Coursework

- Bio-statistics, Robotics, Mechatronics, Biomed Innovation, Tissue/Biomaterial Eng, Control Theory.

University of California, Los Angeles (UCLA)

Los Angeles, CA | Oct 2018 - Mar 2022

Bachelor of Science in Honors Integrative Biology and Physiology GPA: 3.5/4.0, Upper Div 3.84/4.0

- Awards: Department Honors, Alumni Scholarship National Finalist, Achievement Scholarship.

EXPERIENCE

ForUs (Generator Health)

Remote

Operations Specialist

Feb 2026 - Present

- Resolved 100+ daily complex pharmacy cases by coordinating with insurers, providers, and patients.
- Partnered with engineering to improve CRM workflows, leveraging Python/Excel automation to reduce operational bottlenecks and support AI-driven processes.
- Trained team members on prior authorization workflows and tools. Collaborated on drafting new SOP.

Shape Memory Medical Inc.

San Jose, CA

Engineering Intern

Jun 2023 - Aug 2023

- R&D: Designed patient-specific cardiac aneurysm CAD models, 3D-printed prototypes, and integrated with a pulsatile flow system for pre-clinical testing; supported device delivery and data collection.
- Manufacturing: Upgraded Crimping Station circuits, enhancing usability and operational efficiency.
- Quality: Optimized laser welding parameters via Taguchi DOE; achieved 367% tensile strength boost.

Samuel Sia Laboratory - Microfluidics

NY, NY

Graduate Research

Aug 2022 - Dec 2022

- Executed PCR testing and characterized microfluidic pump designs to optimize automated wastewater SARS-CoV-2 surveillance, streamlining sample processing and accelerating early detection.

Colwell Laboratory of Circadian and Sleep Medicine

Los Angeles, CA

Undergraduate Research Assistant

Jul 2019 - Jun 2022

- Led 3-year research on sex- and genotype-dependent neurodevelopment in Bloc-1-deficient Pallid mice. Conducted behavioral experiments, statistical analysis, and presented findings at UCLA (2022).
- Authored and defended a 20-page research paper, earning UCLA Departmental Honors.

Silk Road Medical Inc. (Acquired by Boston Scientific)

Minneapolis, MN

R&D Engineer Intern

Jun 2021 - Sep 2021

- Designed and implemented a microcontroller-based carotid artery test-bed to validate catheter pressure and flow performance; also served as a procedural model for clinician feedback.
- Automated FDA MAUDE root-cause and market analyses, guiding strategic product development.

PROJECTS (kevinod.com/Projects)

- AI deep learning models for malaria cell detection (object detection), breast cancer survival-based treatment using genetic profiles, and cardiovascular risk prediction with subgroup-aware modeling.
- 5-DOF robotic arm utilizing ROS 2 framework, MoveIt 2 path planning, and Gazebo simulation.
- Full-stack, API-based medication reminder system for hospital EHR integration. UCLA YTP team.
- Robotic pull-wire catheter A* Planning & Control algorithm for autonomous vascular navigation.
- Endotracheal Intubation Emergency Model with Functional, Verification, and Validation requirements.
- Autonomous bipedal robot built from CAD, 3D printing, mechatronics, Python, and simulation.

SKILLS

Tech: 3D Printing, Design Control, Mechatronics, Robotics, Laser Cutting, PCR, Experimental Design.

Software: CAD (SolidWorks), Python, AI/ML, ROS 2, Gazebo, MoveIt 2, Data Analysis, Matlab, Excel.